

1. (70 points) Given initial conditions $x(0) = 1$, $x'(0) = -1$, consider the following non-homogeneous linear constant coefficient differential equation

$$x'' + 5x' + 4x = e^t$$

- a. (35 pts) By using the Laplace transform based solution technique, find its solution.
 b. (35 pts) By transforming it to a first-order system of two linear differential equations, solve it using the Laplace transform based solution technique on the obtained linear system.

Do not write over this line!

$$\begin{aligned} \underline{A(a)} \quad & (s^2 \underline{X}(s) - s x(0) - x'(0)) + 5s \underline{X}(s) - 5x(0) + 4 \underline{X}(s) \\ & = \frac{1}{(s-1)} \end{aligned}$$

$$\begin{aligned} \Rightarrow (s^2 + 5s + 4) \underline{X}(s) &= \frac{1}{(s-1)} + s x(0) + (x'(0) + 5x(0)) \\ &= \frac{1}{(s-1)} + s + 4 = \frac{s^2 + 3s - 3}{(s-1)} \end{aligned}$$

$$\underline{X}(s) = \frac{s^2 + 3s - 3}{(s-1)(s^2 + 5s + 4)} = \frac{(s^2 + 3s - 3)}{(s-1)(s+1)(s+4)}$$

$$\underline{X}(s) = \frac{A}{(s-1)} + \frac{B}{(s+1)} + \frac{C}{(s+4)} = \frac{1/10}{(s-1)} + \frac{5/6}{(s+1)} + \frac{1/15}{(s+4)}$$

$$\Rightarrow x(t) = \left[\frac{1}{10} e^t + \frac{5}{6} e^{-t} + \frac{1}{15} e^{-4t} \right] u(t).$$

Do not write over this line!

$$\underline{A}(b) \quad \begin{aligned} x_1(t) &= x(t) \\ x_2(t) &= x'(t) \end{aligned}, \quad \underline{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}, \quad \underline{x}(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$$\underline{x}' = \begin{bmatrix} x_1'(t) \\ x_2'(t) \end{bmatrix} = \begin{bmatrix} x_1'(t) \\ x_2''(t) \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix}}_{\underline{A}} \underbrace{\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}}_{\underline{x}} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\underline{f}(t)} e^t$$

$$\underline{x}' = \underline{A} \underline{x} + \underline{f}(t).$$

$$s \underline{X} - \underline{x}(0) = \underline{A} \underline{X} + \underline{F}(s)$$

$$(s \underline{I} - \underline{A}) \underline{X} = \underline{x}(0) + \underline{F}(s).$$

$$\underline{X}(s) = (s \underline{I} - \underline{A})^{-1} (\underline{x}(0) + \underline{F}(s))$$

$$= \begin{bmatrix} s & -1 \\ 4 & s+5 \end{bmatrix}^{-1} \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s-1} \right)$$

$$= \frac{\begin{bmatrix} s+5 & 1 \\ -4 & s \end{bmatrix}}{(s^2+5s+4)} \begin{bmatrix} 1 \\ \frac{2-s}{s-1} \end{bmatrix} = \begin{bmatrix} s+5 + \frac{2-s}{s-1} \\ -4 + \frac{s(2-s)}{s-1} \end{bmatrix} \frac{1}{(s+4)(s+1)}$$

$$= \begin{bmatrix} \frac{(s^2+3s-3)}{(s-1)(s+1)(s+4)} \\ -\frac{s^2+2s-4}{(s-1)(s+1)(s+4)} \end{bmatrix} = \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix}$$

$$\Rightarrow X_1(s) = \frac{s^2+3s-3}{(s-1)(s+1)(s+4)} \xleftrightarrow{\mathcal{I}^{-1}} x_1(t) = x(t) = \left[\frac{1}{10} e^t + \frac{5}{6} e^{-t} + \frac{1}{15} e^{-4t} \right] u(t)$$

2. (30 points) Given that $e^{2x} \sin x$ is a solution to the following homogeneous 4th order linear constant coefficient differential equation, find its general solution:

$$\frac{d^4 y}{dx^4} - 8 \frac{d^3 y}{dx^3} + 26 \frac{d^2 y}{dx^2} - 40 \frac{dy}{dx} + 25y = 0$$

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(A2) check the characteristic equation:

$$p(s) = s^4 - 8s^3 + 26s^2 - 40s + 25 = 0$$

Given that $e^{2x} \sin x$ is a solution, then $s = 2 + j1$ is a root of the characteristic equation. Therefore $s = 2 - j1$ is also a root (characteristic polynomial is with real coefficients)

$$\begin{aligned} \text{Thus } (s - (2 + j1))(s - (2 - j1)) &= (s - 2)^2 + 1 \\ &= s^2 - 4s + 5 \end{aligned}$$

divides the characteristic polynomial $p(s)$:

$$p(s) = (s^2 - 4s + 5) \cdot (s^2 - 4s + 5) = (s - (2 + j1))^2 (s - (2 - j1))^2$$

Therefore the repeated roots generate the following set of solutions:

$$\{e^{2x} \sin x, e^{2x} \cos x, x e^{2x} \sin x, x e^{2x} \cos x\}$$

Hence, the general solution is:

$$y(x) = \alpha e^{2x} \sin x + \beta x e^{2x} \sin x + \gamma e^{2x} \cos x + \delta x e^{2x} \cos x$$